

AI Demo Knowledge Base

1. React

React is a JavaScript library for building user interfaces. React applications are composed of reusable components. Components can receive data through props and manage changing data with state.

2. FastAPI

FastAPI is a Python framework for building APIs. It supports type hints, automatic request validation, asynchronous endpoints, and automatically generated OpenAPI documentation.

3. RAG

RAG means Retrieval-Augmented Generation. A RAG application retrieves relevant information from a knowledge base before asking a language model to generate an answer. The typical pipeline is: document loading, text chunking, embedding, vector search, context construction, and language model generation.

4. Embedding

An embedding represents text as a numerical vector. Texts with similar meanings can have vectors that are close to each other in vector space. Embeddings are commonly used for semantic search and retrieval systems.

5. Vector Database

A vector database stores vectors and provides similarity search. In an AI knowledge assistant, document chunks can be converted into embeddings and stored in a vector database. When a user asks a question, the system searches for the most relevant chunks.

6. Docker

Docker packages an application and its dependencies into containers. A Docker image is a template used to create containers. Docker Compose can be used to run multiple services together, such as a frontend, backend, database, and Redis.

7. Example Questions

What is React?

What is FastAPI used for?

Explain the RAG pipeline.

What is an embedding?

Why do we need a vector database?

What problem does Docker solve?